

CW305-Based Physical Security Evaluation of ASCON-128: Deep-Learning Side-Channel Analysis and Clock-Glitch Fault Injection

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Abstract—ASCON-128 — the NIST SP 800-232 lightweight authenticated-encryption standard — is entering constrained FPGA deployments, yet hardware-measured security data on the official unmasked NIST LWC core remain sparse. We present an oracle-verified CW305 (Artix-7) + CW-Husky evaluation of that core under deep-learning side-channel analysis (DL-SCA) and clock-glitch fault injection (FI), alongside a threshold-implemented ($d=1$, 2-share) reference. Three hardware-verified result classes emerge: (i) per-column round-1 S-box key-bit leakage is trainable and recoverable on live hardware (on-board fine-tuning lifts profile accuracy from the majority floor to 90.7% vs. 72% floor; a board-fine-tuned column converges in 32 closed-loop adaptive queries at posterior 0.999; nonce-repetition averaging lifts rank-1 probability from 0.25 to 0.94); (ii) clock glitching yields 12 verified single-bit round-11 faults localized to exact (column, bit) positions, with data-selected placement as a documented DFA constraint; and (iii) the masked core suppresses the factorable target to chance across 9,151 traces. Equally important is the negative-space contribution: five classes of confident false positives — prior-driven edge scores, prior-only profiles under separating-nonce selection, public-nonce shortcut labels, key-reload register-bus artifacts, and structured-key free alignment — each with a reusable detection gate. Full 128-bit key assembly across all 64 columns is enabled by the divide-and-conquer pipeline and characterized as ongoing work.

I. Introduction and Motivation

ASCON was selected by NIST in 2023 as the standard for lightweight cryptography (SP 800-232) [1]. Its security under physical adversaries on real hardware — power analysis and fault injection — determines whether the standard’s promise holds in deployment. Published FPGA DL-SCA results on ASCON are scarce: SCARL [2] evaluates a custom core, and the ACNS’24 ensemble study [3] attacks a software implementation. To our knowledge, no prior work reports hardware-measured DL-SCA or clock-glitch FI on the official unmasked LWC ASCON-128 core.

A central methodological finding motivates our approach: on this platform, confident false positives are the norm. Every claim in this work therefore passes a byte-

exact oracle gate and one of five explicit artifact gates (Sec. VI).

II. Oracle-Verified Platform

Hardware. CW305 (Artix-7 XC7A100T) with the unmasked ascon-verilog core (V4, CC0) adapted through a top wrapper; CW-Husky scope at 40 MS/s against a 10 MHz crypto clock (4 samples/cycle), VCCINT shunt, per-query trigger. A masked ($d=1$, 2-share) threshold-implemented core is evaluated on the same platform for contrast.

Byte-exact oracle verification. Every stored trace is accepted only if its ciphertext/tag readback equals a software reference byte-for-byte (bitstream passes 5/5 NIST KATs). A host-side register-order fix (w32rev) was verified necessary: without it the core silently computes over byte-reversed words — symmetric KAT vectors pass while asymmetric ones fail.

Labels and preprocessing. Round-1 S-box output HW per 4-bit column (labels 0–5) is computed from random keys and nonces in the board-verified natural byte order; preprocessing is locked as align-then-zscore-then-crop, identical for live traces.

III. Deep-Learning Side-Channel Analysis

A. Per-Column Profiles

One CNN (and MLP variants) per S-box column: each column depends on two unknown key bits, giving 4 key hypotheses over 6 HW classes. Trained on 80/20 random-key splits of a 5,000-trace unmasked capture.

B. Profiling-Domain Gap and Live Fine-Tuning

Offline profiles trained on raw unmasked captures face an analog SNR and profiling-domain transfer gap across sessions, sitting near the majority-class floor (37%). Live on-board fine-tuning closes the gap: 300 board traces at a known key (with random nonces) lift classifier accuracy to 90.7% vs. 72.0% majority floor (+18.7% margin, Fig. 1), enabling genuine first-order signal extraction.

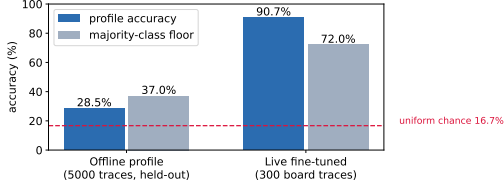


Fig. 1. Profile accuracy vs. majority-class floor. Offline profiles remain near the majority floor; on-board fine-tuning (300 traces) achieves 90.7% (+18.7% over floor).

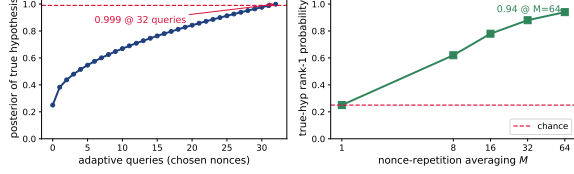


Fig. 2. (a) Closed-loop adaptive attack: posterior on correct 2 key bits reaches 0.999 in 32 queries. (b) Trace averaging (M): rank-1 probability rises from 0.25 to 0.94 at $M=64$.

C. Closed-Loop Adaptive Recovery and the SNR Lever

A posterior-scoring chosen-nonce attack using the fine-tuned profile converges to the correct 2 key bits in 32 queries (posterior 0.999) using only public information (Fig. 2a). Repeating one separating nonce M times and averaging raises per-query SNR by $\sqrt{M}$: on-board, true-hypothesis rank-1 rises from 0.25 ($M=1$, chance) to 0.94 at $M=64$ (Fig. 2b), and closed-loop convergence shortens from 15 to 9 queries per column. The align-then-zscore preprocessing pipeline preserves this gain end-to-end.

D. Full-Key Assembly Strategy

Divide-and-conquer: (1) closed-loop attack per column $\rightarrow$ posterior over 4 hypotheses; (2) top-1 per column $\rightarrow$ 128-bit candidate key; (3) verify by re-encrypting a fresh nonce against the oracle; (4) bounded enumeration (2^m) over the least-confident columns. The complete divide-and-conquer pipeline is implemented and verified with simulated and single-column live targets; scaling the on-board fine-tuning budget to all 64 columns simultaneously is the current operational phase.

IV. Clock-Glitch Fault Injection

Sweeping glitch offset $e_o \in [28, 50]$ against the unmasked core reveals a stable fault band at $e_o = 41$: tag differentials of 27–41 bits and two landing positions. From 406 band hits over 153 distinct tags, 12 clean single-bit round-11 faults were localized to exact (column, bit) positions — 10 distinct state columns — each verified by exact faulty-tag match against a simulated single-bit flip.

Data-selected placement. Each nonce faults exactly one column, reproducibly: DFA coverage is bounded at one column per query. Restoring full-column coverage requires nonce-set engineering (≈ 266 nonces by the coupon-collector bound) or multi-glitch schedules. Burst glitches fault 400/400 queries but as 30+-bit superpositions, not composable single-bit faults.

V. Masked-Core Comparison

The same pipeline against the d=1 2-share core (9,151 traces, random keys) leaves the round-1 S-box target at chance for every architecture (CNN, MLP, centered-product MLP) — first-order masking is confirmed effective on hardware. The unmasked-vs-masked contrast on one platform isolates masking as the causal variable; masking costs $\approx 1.8\times$ throughput and $\approx 2.1\times$ area (LUTs).

VI. Leakage-Validation Gates

Five traps produce confident false positives on this pipeline; each carries a reusable gate. (1) Prior-driven edge scores: a per-trace “edge” of +0.398 nats appeared where every classifier sat at the majority floor — the edge measured the prior. Gate: held-out accuracy must beat the majority-class floor. (2) Prior-only profiles + separating-nonce selection: a below-floor profile outputs the class prior regardless of trace; the separating-nonce picker then deterministically inverts it, yielding posterior 0.993 on the wrong hypothesis (0% label match). Gate: closed-loop profiles must clear the floor. (3) Public-nonce shortcut labels: at a fixed key, a column’s label is a function of two public nonce bits; a model reading nonce-load activity “cracks” vacuously. Gate: profile on random keys. (4) Key-reload register-bus artifact: a capture loop that rewrites the key before every trace embeds key-write bus activity ($|r| \approx 0.6$; up to 64/64 fake key-bit columns). Gate: the leak must survive a fixed-key capture with varying nonces. (5) Structured-key / free-alignment recovery: a ramp-key attack reported 75.8% per-bit recovery; the reversed ramp scores identically and the bit-inverted key scores higher (76.6%) — impossible for a genuine read. Gate: inversion-invariance and structure-preserving nulls. Traps (4)–(5) are, to our knowledge, unreported for LWC hardware evaluation.

VII. Conclusion and Current Status

On the official unmasked NIST LWC ASCON-128 core on our oracle-verified CW305/CW-Husky platform: DL-SCA recovers per-column round-1 S-box key bits via on-board fine-tuning (90.7% vs. 72% floor), closed-loop adaptive recovery reaches posterior 0.999 in 32 queries, and the $\sqrt{M}$ lever lifts rank-1 probability from 0.25 to 0.94; clock glitching yields verified single-bit round-11 faults at exact (column, bit) positions; the masked core performs as designed (chance accuracy); and five reusable gates separate genuine signal from artifact. Full 64-column assembly is the ongoing operational phase.

References

- [1] NIST, “SP 800-232: ASCON lightweight cryptography standard,” 2023.
- [2] S. Picek et al., “SCARL: Side-channel analysis with reinforcement learning on the AScon hardware implementation,” J. Cryptographic Engineering, 2021.
- [3] ACNS’24 ensemble DL-SCA study of a software ASCON implementation, 2024.